Forecasting Disease Outbreak Reporting Across Indian States: An Uncertainty-Aware Test of Generalization

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***Abstract*—Before the modelling problem of building an early-warning system for infectious disease in India, there is a data problem. Detailed health records are not public, and the only open weekly source, the outbreak reports of the Integrated Disease Surveillance Programme (IDSP), records whether an outbreak was *reported*, not how many people fell ill. We accept that constraint and ask a narrower question: can one pipeline predict whether a state will report an outbreak one to four weeks ahead, and does it still work when the state or the disease changes? We use the EpiClim dataset (8,985 records, 2009–2022), aggregated to state and week, with climate rebuilt from NASA POWER, and test five model families in a time-ordered protocol that checks for data leakage. A result counts only if the lower bound of its 95% bootstrap confidence interval for PR-AUC exceeds the base rate. Dengue clears this bar in 31 of 32 results across Maharashtra and Kerala, whereas acute diarrhoeal disease clears it in only 1 of 80 results across five states. Climate adds at most 0.014 PR-AUC and accuracy is barely affected by lead time, so the models have learned the seasonal pattern of reporting.**

***Index Terms*—disease surveillance, outbreak forecasting, generalization, dengue, gradient boosting, SARIMA, explainable AI**

I. Introduction

Indian states experience outbreaks of dengue and diarrhoea every year, and a warning even a few weeks in advance can help a district prepare staff, test kits and mosquito control. AI has often been suggested for this. However, a recent scoping review of AI-based early-warning systems (EWSs) [1] found that they have a high requirement for data quality, are hard to inspect and validate, and are rarely tested outside the place where they were developed.

In India, what can be forecast depends on what data can be downloaded. Only health personnel have access to the case-based surveillance platform, and yearly state totals are too coarse to be used on a weekly basis. What remains is the IDSP weekly outbreak report [2], which EpiClim [3], [4] converted into a district-wise weekly table. A row is present only when a report is filed, so an empty week may mean that there was no outbreak or that none was reported. We therefore predict *reporting*, not incidence.

We make three hypotheses before testing: climate data improves forecasts over reporting history alone (H1); one pipeline is sufficient for all state–disease combinations (H2); and explainable AI reveals the drivers of forecasted risk (H3). Our work makes three contributions: a formulation of open Indian surveillance data built around the reporting process, with every target date checked for leakage; a test of generalization of one pipeline on seven state–disease combinations, with a fixed acceptance rule based on the confidence interval; and a data quality audit of EpiClim, in which impossible climate values are replaced and the collapse in reporting in 2020 is quantified.

II. Related Work

Lowe *et al.* [5] demonstrated why dengue forecasts should come with a measure of uncertainty, using a prototype early-warning system in Brazil. In India, Patil and Pandya [6] forecast monthly dengue cases in nine cities of Maharashtra, with random forests performing best, and Sophia *et al.* [7] forecast dengue deaths in Pune two months ahead using lagged climate data. Both studies look at one disease in one state, and neither asks whether the method works in other places. EpiClim [3] was used to explore relationships between climate and disease, but not to forecast. For explanations, SHAP [8] and permutation importance [9] often disagree [10], so we use and compare both. The gaps in our two base papers connect to this work as follows. Generalization is rarely tested [1], so we test one pipeline on seven state–disease combinations with a

fixed acceptance rule. For data quality [1], the source climate is audited and rebuilt, and the 2020 disruption is excluded. For transparency [1], SHAP is cross-checked with permutation importance. There is no forecasting benchmark [3], so we provide a time-ordered benchmark of five model families at four horizons. On reporting vs. occurrence [3], the target is defined as reporting and our claims are limited to it.

III. Data and Method

A. Data

EpiClim holds 8,985 outbreak records from 4 May 2009 to 29 June 2022. District data was too sparse to use: even Nadia in West Bengal, the district that reported dengue most often, has only 29 report weeks in 14 years. We therefore aggregated to state and ISO week on a full grid of 731 weeks. The target $y_{s,d,t}$ is 1 if any outbreak of disease d was reported in state s in week t, and 0 otherwise. Variant names for the same disease were merged, and 1,098 reports from the same week were summed into 497 groups. The source climate contained impossible values, such as $-11.85\,°\text{C}$ in Karbi Anglong, Assam, so we rebuilt weekly temperature, rainfall and humidity from NASA POWER [11] at the centre of each state. From 2019 to 2020, reporting dropped by 50–75% in every state and disease at the same time (Karnataka dengue: from 42.3% to 11.3% of weeks), which points to disruption from COVID-19, so 2020 is excluded. After series with too few reports are dropped, seven combinations remain, with these test base rates: Maharashtra dengue 0.333 and Kerala dengue 0.231; and for acute diarrhoeal disease (ADD), Karnataka 0.269–0.295 (varying with horizon), Maharashtra 0.141, West Bengal 0.385, Gujarat 0.244 and Odisha 0.244.

B. Method

For horizon $h \in \{1, 2, 3, 4\}$ weeks, a model estimates $P(y_{s,d,t+h} = 1 \mid \mathbf{x}_{s,d,t})$ using: whether there was a report in each of the last 4 weeks; the 4- and 8-week rolling reporting rates (the share of recent weeks with a report); week-of-year sine and cosine terms; climate from the current week and the 4 weeks before; a COVID flag; and state and disease indicators. We shift the target forward in time, never the features, and our code checks that every target week starts exactly $7h$ days after its feature week. Data are split by time and never shuffled: training 2009–2017, validation 2018–2019, and test from 1 January 2021 to 29 June 2022 (78 weeks).

We compare persistence and seasonal baselines, logistic regression, XGBoost [12] (depth 3, 200 trees, learning rate 0.05, early stopping on validation PR-AUC), SARIMA [13] (Fourier terms with period 52, and no climate), and stochastic gradient boosting [14] (subsample 0.8, balanced weights). Most weeks have no report, so a model that always predicts "no report" would look accurate; we therefore score with PR-AUC [15], where a random ranking scores the base rate π. For each result, a 95% interval is computed from 1,000 bootstrap resamples [16], and a result is *above chance* only if the lower bound of this interval exceeds π. With four models and four horizons, each combination has 16 results.

IV. Results

Dengue. In Maharashtra, all 16 results clear chance. On the test set (base rate 0.333, 26 positive weeks of 78), the PR-AUC [95% bootstrap CI] at horizons of 1, 2, 3 and 4 weeks is 0.639 [0.469, 0.857], 0.692 [0.513, 0.864], 0.717 [0.530, 0.872] and 0.568 [0.397, 0.786] for XGBoost; 0.643 [0.461, 0.831], 0.618 [0.456, 0.839], 0.625 [0.451, 0.831] and 0.637 [0.449, 0.824] for logistic regression; 0.575 [0.412, 0.798], 0.555 [0.408, 0.783], 0.591 [0.418, 0.780] and 0.596 [0.398, 0.775] for SARIMA; and 0.624 [0.459, 0.819], 0.683 [0.508, 0.849], 0.644 [0.472, 0.845] and 0.625 [0.427, 0.804] for stochastic gradient boosting. The lowest lower bound is 0.397 against a base rate of 0.333, and the point estimates range from 0.555 to 0.717, about double the base rate. There is no clear winner, however, because the intervals of all four models overlap at every horizon. Kerala has half as many positive training weeks as Maharashtra (68 vs. 138), yet 15 of its 16 results clear chance. The only miss is logistic regression at h=4, which falls short by just 0.001 (lower bound 0.230 vs. base rate 0.231). Kerala's best result is stochastic gradient boosting at h=1 (0.689).

Diarrhoeal disease. Only one of the 80 ADD results clears chance: XGBoost at h=1 in Karnataka, with 0.485 [0.311, 0.661] against a base rate of 0.295, decaying to 0.287 at h=4. None of Maharashtra, West Bengal, Gujarat or Odisha has a result above chance.

Climate and lead time (H1). Adding climate raised PR-AUC by only $+0.014$ for logistic regression and $+0.001$ for XGBoost, and SARIMA, which uses no climate at all, stays close to the machine learning models. In Maharashtra, logistic regression scores 0.643, 0.618, 0.625 and 0.637 at horizons 1–4. If the models were acting on short-term momentum, they would become less accurate the further ahead they looked, but they do not. The week of the year alone is also not enough: the seasonal baseline dropped from 0.575 PR-AUC in training to 0.200 in test.

Explanations (H3). SHAP ranks the 8-week rolling reporting rate as the most important feature (mean $|\text{SHAP}|$ 0.129), but permutation importance does not match: most features score near zero, and the same 8-week rate scores $+0.11$ for two combinations but -0.059 for Maharashtra ADD.

V. Discussion and Limitations

The results agree. Climate adds almost nothing, a climate-free SARIMA keeps up with the other models, and accuracy stays flat as the horizon grows. The models have therefore learned *when* each state usually reports dengue, with reports peaking in ISO week 20 in Maharashtra and week 21 in Kerala. This is helpful for planning for the season, but it is not early detection of an unusual outbreak. Our best explanation for the failure on ADD is that diarrhoeal outbreaks occur throughout the year and are caused by local events, such as a contaminated water supply, that a state-level series cannot pick up; our data cannot test this directly. We conclude that H1 is not supported, that H2 holds for dengue but not across diseases, and that H3

is not confirmed, so we do not treat SHAP output as evidence of what drives outbreaks.

We do not claim to be more accurate than previous dengue studies [6], [7], which predict monthly cases or deaths that cannot be compared with PR-AUC on weekly reporting. Our contribution is the test of generalization that the scoping review [1] called for, run on a corrected version of EpiClim [3], with the negative ADD result reported in full.

The main limitation is that we forecast reporting, not infection. Climate comes from one location per state. The 78-week test set gives wide intervals. Only two states could be tested for dengue, and reporting stayed low after 2020 in some test series. Finally, the scores rank weeks by risk and are not calibrated probabilities.

VI. Conclusion

The leakage-checked pipeline predicted dengue outbreak reporting above chance in 31 of 32 results across two states, but diarrhoeal disease reporting in only 1 of 80 results across five states. The signal comes from the seasonal pattern of reporting, not from climate. Next, we will add IDSP reports from July 2022 to 2026, verify them against EpiClim where the two overlap, test the trained models out of time on these new years, and compare reporting with yearly dengue case counts. The code and a live dashboard are available at https://github.com/sumukh-dgaf/disease-outbreak-ai.

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